Kibum Kim

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Research Overview

Ph.D. candidate at KAIST (advisor: Prof. Chanyoung Park), graduating in July 2027. I aim to build a multimodal AI assistant for wearable devices that understands a scene in detail and still responds in real time.

- **Fine-grained visual understanding:** scene graph generation under long-tailed and weak supervision [2, 4, 6, 10], compositional reasoning in vision-language models [16, 24], and gaze-guided egocentric video understanding [22].
- **Efficient real-time inference:** visual token pruning and knowledge distillation [16, 18, 19], applied to smart glasses at Huawei R&D London [21, 22].
- **Current work:** proactive AI assistants and test-time scaling for video understanding.

Education

Jul 2023 – Jul 2027 (expected) — **Ph.D., KAIST** — Daejeon, South Korea
Research interests: efficient vision-language models, real-time video understanding, scene understanding. Advisor: Prof. Chanyoung Park.

Jul 2021 – Jul 2023 — **M.S., KAIST** — Daejeon, South Korea
Research focus: scene understanding, recommender systems. Advisor: Prof. Chanyoung Park

Research Experience

Mar 2026 – Sep 2026 — **Research Intern, Huawei R&D London** — London, UK
Smart-glasses team. Host: Zhensong Zhang, Principal Researcher.

- **Streaming video MLLMs:** Developed visual token compression for real-time, on-device inference on smart glasses; kept only 5% of visual tokens and cut latency 6× with a 1.7% accuracy drop [21].
- **Gaze-aware egocentric QA:** Trained a model to learn when to trust the wearer's eye gaze for frame selection, gaining 5.2 points on a wearer-authored test set [22].
- **Demo and challenge:** Built an end-to-end smart-glasses demo, and placed 2nd in Meta's EgoLongQA Challenge at ECCV'26.

Aug 2025 – Feb 2026 — **Visiting Scholar, University of California San Diego** — San Diego, USA
Host: Prof. Julian McAuley.

- **Efficient video LLMs:** Proposed visual token pruning that keeps fine-grained details at lower inference cost [18]; Best paper honorable mention at the ECCV'26 Wearable AI Workshop.
- **LLM recommendation:** Represented items with images to make LLM recommender systems token-efficient, in collaboration with Amazon [15].

Publications

* equal contribution

[24] **AAAI'27** under review — **Kibum Kim**, Kyle Min, Eduardo Escoto, Yueqi Wang, Julian McAuley, Chanyoung Park. *When Data Scaling isn't Enough: Leveraging Video Scene Graphs to Improve Compositional Reasoning in Video-Language Models.*

[23] **ICLR'27** under review — Jiwan Kim, **Kibum Kim**, Byung-Kwan Lee, Chanyoung Park. *Easy to Call, Hard to Use Effectively: Learning How and When to Use Visual Tools in Small MLLM Agents.*

[22] **ICLR'27** under review — **Kibum Kim**, Weitong Cai, Jiankang Deng, Songcen Xu, Shan Gao, Dehua Song, Jifei Song, Jiwan Kim, Hongseok Kang, Hyunjun Jung, Chanyoung Park, Hang Zhang, Zhensong Zhang. *When to Trust the Eyes: Learning Adaptive Gaze Reliance in Egocentric Video.*

[21] **ICLR'27** under review — **Kibum Kim**, Dingli Liang, Yukai Huang, Haozheng Zhang, Weitong Cai, Rongguang Wang, Xiang Chen, Min Liu, Renjiu Hu, Gaolei Li, Yiqiao Xie, Jifei Song, Chanyoung Park, Zhensong Zhang, Hang Zhang. *Structure-Preserving Visual Token Compression for Streaming Video Understanding via Bilateral Grid Splatting.*

[20] **WACV'27** SeonWoo Choi, **Kibum Kim**, Chanyoung Park, Donghyun Kim. *Personalized Visual Representation Alignment for Generative Multimodal Recommendation.*

[19] **ECCV'26** Jiwan Kim, **Kibum Kim**, Wonjoong Kim, Byung-Kwan Lee, Chanyoung Park. *Why and When Visual Token Pruning Fails? A Study on Relevant Visual Information Shift in MLLMs Decoding.*

[18] **ECCV'26** **Kibum Kim**, Jiwan Kim, Kyle Min, Yueqi Wang, Jinyoung Moon, Julian McAuley, Chanyoung Park. *Sink-Token-Aware Pruning for Fine-Grained Video Understanding in Efficient Video LLMs.*

[17] **ICLR'26** Heewoong Noh, Namkyeong Lee, Gyoung S. Na, **Kibum Kim**, Chanyoung Park. *IR-Agent: Expert-Inspired LLM Agents for Structure Elucidation from Infrared Spectra.*

[16] **ICLR'26** Jiwan Kim, **Kibum Kim**, Sangwoo Seo, Chanyoung Park. *CompoDistill: Attention Distillation for Compositional Reasoning in Multimodal LLMs.*

[15] **ICLR'26** **Kibum Kim**, Sein Kim, Hongseok Kang, Jiwan Kim, Heewoong Noh, Yeonjun In, Kanghoon Yoon, Jinoh Oh, Julian McAuley, Chanyoung Park. *Token-Efficient Item Representation via Images for LLM Recommender Systems.*

[14] **NeurIPS'25** Yeonjun In, Kanghoon Yoon, Sukwon Yun, **Kibum Kim**, Sungchul Kim, Chanyoung Park. *Training Robust Graph Neural Networks by Modeling Noise Dependencies.*

[13] **EMNLP'25** Findings Yeonjun In, Wonjoong Kim, Kanghoon Yoon, Sungchul Kim, Mehrab Tanjim, **Kibum Kim**, Chanyoung Park. *Is Safety Standard Same for Everyone? User-Specific Safety Evaluation of Large Language Models.*

[12] **KDD'25** Sein Kim*, Hongseok Kang*, **Kibum Kim**, Jiwan Kim, Donghyun Kim, Minchul Yang, Kwangjin Oh, Julian McAuley, Chanyoung Park. *Lost in Sequence: Do Large Language Models Understand Sequential Recommendation?*

[11] **SIGIR'25** Jiwan Kim, Hongseok Kang, Sein Kim, **Kibum Kim**, Chanyoung Park. *Disentangling and Generating Modalities for Recommendation in Missing Modality Scenarios.*

[10] **ICLR'25** **Kibum Kim**, Kanghoon Yoon, Yeonjun In, Jaehyeong Jeon, Jinyoung Moon, Donghyun Kim, Chanyoung Park. *Weakly Supervised Video Scene Graph Generation with Natural Language Supervision.*

[9] **AAAI'25** Kanghoon Yoon, **Kibum Kim**, Jaehyeong Jeon, Yeonjun In, Donghyun Kim, Chanyoung Park. *ReTAG: Retrieval-Augmented Scene Graph Generation via Multi-Prototype Learning.*

[8] **CIKM'24** Kanghoon Yoon, Yeonjun In, Namkyeong Lee, **Kibum Kim**, Chanyoung Park. *Debiased Graph Poisoning Attack via Contrastive Surrogate Objective.*

[7] **ECCV'24** Jaehyeong Jeon, **Kibum Kim**, Kanghoon Yoon, Chanyoung Park. *Semantic Diversity-aware Prototype-based Learning for Unbiased Scene Graph Generation.*

[6] **CVPR'24** **Kibum Kim**, Kanghoon Yoon, Jaehyeong Jeon, Yeonjun In, Jinyoung Moon, Donghyun Kim, Chanyoung Park. *LLM4SGG: Large Language Model for Weakly Supervised Scene Graph Generation.*

[5] **WWW'24** **Oral** Yeonjun In, Kanghoon Yoon, **Kibum Kim**, Kijung Shin, Chanyoung Park. *Self-guided Robust Graph Structure Refinement.*

[4] **ICLR'24** **Kibum Kim***, Kanghoon Yoon*, Yeonjun In, Jinyoung Moon, Donghyun Kim, Chanyoung Park. *Adaptive Self-training Framework for Fine-grained Scene Graph Generation.*

[3] **SIGIR'23** **Kibum Kim**, Dongmin Hyun, Sukwon Yun, Chanyoung Park. *MELT: Mutual Enhancement of Long-Tailed User and Item for Sequential Recommendation.*

[2] **AAAI'23** Kanghoon Yoon*, **Kibum Kim***, Jinyoung Moon, Chanyoung Park. *Unbiased Heterogeneous Scene Graph Generation with Relation-aware Message Passing Neural Network.*

[1] **CIKM'22** Sukwon Yun, **Kibum Kim**, Kanghoon Yoon, Chanyoung Park. *LTE4G: Long-Tail Experts for Graph Neural Networks.*

Workshop papers

[W2] **ECCV'26** **Oral** **Kibum Kim**, Jiwan Kim, Kyle Min, Yueqi Wang, Jinyoung Moon, Julian McAuley, Chanyoung Park. *Sink-Token-Aware Pruning for Fine-Grained Video Understanding in Efficient Video LLMs.* ECCV 2026 Wearable AI Workshop.

[W1] **NeurIPS'25** **Spotlight** **Kibum Kim**, Kyle Min, Chanyoung Park. *Data Scaling isn't Enough: Towards Improving Compositional Reasoning in Video-Language Models.* Efficient Reasoning Workshop.

Honors and Awards

2026	2nd Place, EgoLongQA Challenge (small track), ECCV'26 Wearable AI Workshop, Meta
2026	Best Paper Honorable Mention, ECCV'26 Wearable AI Workshop, Meta
2026	Outstanding Reviewer, CVPR'26
2026	Top Reviewer (Gold), ICML'26
2022	Poster Competition Excellence Award, Department of ISysE, KAIST
2022	Hanyang Academic Achievement Award, Hanyang University

Leading Projects

Dec 2025 – Present	**Foundation Model for Electromyography (EMG) in the Medical Domain** With SNU College of Medicine (Prof. Woo-Hyung Lee). Built end-to-end medical AI pipelines covering dataset preparation, foundation model training, and standardized evaluation protocols.
Jul 2021 – Present	**Visual Intelligence Technique Development** With ETRI. Fine-grained relational scene understanding in images and videos [2, 4, 6, 7, 9, 10], aiming to build long-term memory in video via scene graphs.
Jul 2022 – May 2026	**AI for Reasoning, Extraction, and Understanding of Commonsense** With IITP. Visual commonsense reasoning in images and videos using deep scene understanding.
Jun 2024 – Jul 2025	**NAVER & Intel & KAIST (NIK) AI Collaboration Project** With NAVER and Intel. Expanded the Intel Gaudi AI ecosystem by developing and open-sourcing multimodal LLM-based recommender systems.

Research Collaborations

Industry	**Oracle** (Kyle Min, Principal Applied Scientist) Aug 2025 – Mar 2026 Efficient video understanding. Led the work as first author: ECCV'26 [18], AAAI'27 under review [24], NeurIPS'25 workshop spotlight [W1]. **Amazon** (Jinoh Oh, Applied Scientist) Dec 2024 – Aug 2025 Efficient multimodal LLM-based recommendation. Led the work as first author: ICLR'26 [15].
Academia	**Korea University** (Prof. Donghyun Kim) 2023 – Present Scene graph generation [4, 6, 9, 10] and generative multimodal recommendation [20]. **Seoul National University College of Medicine** (Prof. Woo-Hyung Lee) Dec 2025 – Present Foundation models for electromyography (EMG) signals in the medical domain (ongoing).

Professional Service

Conference reviewer	CVPR, ECCV, ICML, ICLR, NeurIPS (D&B), AAAI, KDD, WWW, CIKM (2024–2027)
Workshop reviewer	Efficient Large Vision Models (CVPR'25), Multimodal Algorithmic Reasoning (NeurIPS'25), Efficient Reasoning (NeurIPS'25), Towards Robots with Human-Level Abilities (ICLR'25), Graph Learning with Foundation Models (PAKDD'25)
Journal reviewer	TKDE, TNNLS, Pattern Recognition Letters, Neurocomputing, Knowledge-Based Systems, Journal of Big Data

Technical Skills

Programming	Python, PyTorch, Hugging Face Transformers, Git, Linux
Models	Token pruning/compression, knowledge distillation, LoRA fine-tuning & RL post-training [23]
Data	LLM-driven data curation and label generation [6,10,13,22]

References

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